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1 1) Core choice: what is an "area quantum"?

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1.1 👤 You

Please search for knowledge related "Principal Differential Geometric Algebra" and "compute using Grassmann.jl, Cartan.jl" as explained in a book of Michael Reed and in the github project under https://github.com/chakravala.

Question 1: Are you aware in the sense of "ready to apply" of all math related to "Differential Geoemtri Algebra"?

Question 2: Can you apply this the foundations of Burkhard Heims theory?

1.2 🤖 ChatGPT

I'll look up the relevant book/project context (Michael Reed + chakravala / Grassmann.jl / Cartan.jl), then answer both questions directly in terms of what I can apply now and what would still need formalization.

I found the key references and package claims. Next I'll verify a concise description of Burkhard Heim's theory context so I can answer Q2 precisely.

Thought for 17s

Yes — I'm now oriented to the exact ecosystem you mean:

Grassmann.jl is presented as a Grassmann–Clifford–Hodge multilinear differential geometric algebra framework with exterior/regressive/inner/geometric products, Hodge star, boundary operators, and compile-time tensor-algebra typing via DirectSum.jl / AbstractTensors.jl. GitHub

Cartan.jl is presented as a Julia framework for tensor fields / frame bundles / manifolds, aiming at executable differential geometric algebra and geometric PDEs on grids/meshes/simplicial decompositions, with TensorField constructions and references to Reed's DGA texts/books. GitHub

The GitHub/Makie issue confirms the 2025/2026 "Principal Differential Geometric Algebra: compute using Grassmann.jl, Cartan.jl" book context and its relation to Grassmann/Cartan/-Makie examples. GitHub

Q1

No, not in the literal sense of "all math, ready-to-apply" for every statement in that body of work.

What I am ready to apply (at a useful level):

exterior algebra / Clifford (geometric) algebra identities differential forms, Hodge duality, wedge/interior products

connections, curvature forms, Lie derivatives/brackets (standard formulations)

manifold/tangent-bundle coordinate constructions

translation of many of these objects into Grassmann.jl / Cartan.jl-style computational workflows

What is not safe to claim without your concrete target chapter/formulation:

exact coverage of all Reed-specific definitions, conventions, and notation choices

every package-specific API nuance/version detail

every derivation in a 545-page book (especially if it introduces nonstandard terminology or custom identities)

So: ready to work with it seriously, yes; omniscient over all details, no.

Q2

Yes, I can help apply differential geometric algebra machinery to the foundational mathematics of Burkhard Heim-related formulations — but with an important condition:

I need a precise formalization of which “Heim theory” layer you mean (original Heim, later extensions, a specific author’s reconstruction, mass formula sector, geometric axioms, etc.).

Heim theory is not mainstream / not broadly accepted in contemporary physics, and available descriptions are often mixed with speculative or secondary interpretations. Wikipedia+1

What I can do well:

recast claims into explicit math objects (manifolds, bundles, forms, metrics, connections, operators)

check internal consistency (dimensions, index structure, covariance, algebra closure)

derive a computable prototype in Julia using Grassmann.jl / Cartan.jl

separate:

mathematical formalism

computational implementation

empirical/physical claims (which require independent evidence)

Practical way to proceed

If you want, send one small target (even rough notes), e.g.:

“Heim’s space dimensional decomposition”

“mass operator / metric ansatz”

“field equations in notation X”

“one chapter theorem / derivation from Reed”

and I can convert it into a DG-algebra / Cartan-form formulation plus a Julia skeleton (types, basis, operators, test identities).

Sources

1.3 👤 You

You are right Heim Theory has several components and extensions. I would like to focus on the smallest possible scale, where Heim defines some area quanta, these elements are clearly defined by Heim and most extension do not cover their details. But to develop Heim's theory we should start with such area quants and their detailed description. Are these rectangular, triangle shape or hexagonal. Are they bi-vectors and if yes in what space. I would like to assume rigid elements whose area/size might change over time. Heim had discussed some spheres as a starting point in time and then these spheres expand. Some sort energy function related to the curvature of these spheres is a global energy balance. We can assume that these elements are very small (Planck length). We can assume that there is an energy related to angles between such elements (only the hinges can freely move). Such a geometric structure has already a big number of constraints (3-diagonal like spare matrix form). At this stage we have no fields, no particle and no possibility to generate some ad hoc curvature. To have some bending at one place we have to move a infinite amount of elements or we have to assume a curved boundary condition. Please try to generate a theory according to Heim ideas build from the area quanta. Assume that movement of curved ensemble take no energy, but their is causality speed limit. To move such an ensemble, you have to bend some elements in front of the structure and relax some elements at the end of the structure. Please keep in mind that quantum theory's wave function and the square of its value might be discovered as simple functions of curvature, where the square values are related to energy and mass.

1.4 🤖 ChatGPT

I'll build a minimal geometric model from area quanta only (kinematics + constraints + energy + causal update rule), then show how a wavefunction-like quantity can emerge from curvature amplitudes.

Thought for a few seconds

Excellent direction. Starting from area quanta (not particles/fields) is the right "smallest-scale" move if the goal is to make Heim-like ideas mathematically testable.

Below is a minimal constructive theory you can actually develop. It is not a claim about Heim's exact original equations; it is a Heim-inspired foundation built from area elements, hinges, curvature, and causal propagation.

1) Core choice: what is an "area quantum"? Recommendation

Treat each area quantum as a bivector-valued oriented plaquette:

geometrically: a small rigid 2-cell (triangle is best)

algebraically: a bivector $B \in \Lambda^2(T_x M)$

metric size: $|B| \sim \ell_P^2$ (Planck-area scale)

shape: fixed combinatorial type, variable metric embedding

Why triangles (best minimal choice)

Triangles are topologically stable building blocks for arbitrary surfaces/manifolds (simplicial complex).

Local curvature at hinges is standard (Regge-like).

No ambiguity in planarity of a 2-cell.

Hexagons/rectangles are convenient for lattices, but triangles are universal.

Alternative choices

Rectangles: easier indexing, but less natural under arbitrary curvature (shear ambiguity).

Hexagons: good for near-isotropic tilings in 2D, but not universal simplicial primitives.

Conclusion: use triangular plaquettes as primitive, allow coarse-grained hexagonal patterns to emerge.

2) What space are the bivectors in?

This is crucial.

Minimal answer

Each area quantum is a bivector in the local tangent space:

$B_f \in \Lambda^2(T_x M)$

where f labels a face (triangle) and x is its location (or cell center).

If you want a stricter algebraic implementation:

choose a local frame e_a,

represent

$B_f = \frac{1}{2} B_f^{ab} e_a \wedge e_b$. Which dimension?

You can keep this general at first:

Kinematic model: M is a 3D spatial manifold and time is an update parameter.

Covariant model: M is 4D spacetime and plaquettes may be spacelike/timelike.

Heim-inspired extension: if you later need extra dimensions, you can enlarge $T_x M$ to n-D without changing the plaquette machinery.

Best starting point: Use 3D space + discrete time. Then each area quantum is a spatial bivector in $\Lambda^2(\mathbb{R}^3)$, which is dual to a normal vector. This makes intuition easy.

3) Minimal kinematics: rigid faces, flexible hinges

Your idea is strong: rigid elements + hinge motion.

Define a simplicial complex K with:

vertices V

edges E

triangular faces F

Each face $f \in F$:

has fixed rest area $A_{0,f} \sim \ell_P^2$ (or nearly fixed) carries orientation and bivector B_f

may allow small area dilation $A_f(t)$

Each pair of adjacent faces sharing edge e:

has a dihedral angle θ_e

This gives a discrete geometry with no continuous metric field assumed.

4) Curvature from hinges (discrete and local)

This is the natural place to define curvature without "ad hoc curvature."

Hinge curvature variable

For each edge e, define a curvature/bending variable

$\kappa_e \equiv \theta_e - \theta_{0,e}$

where $\theta_{0,e}$ is the reference angle (flat or vacuum value).

$\kappa_e = 0$: local vacuum shape

$\kappa_e \neq 0$: stored bending / curvature excitation

This is exactly the right object if the only free motions are hinge motions.

5) Energy functional (local, geometric, no fields yet)

You asked for:

area/size may change over time

energy from curvature of expanding spheres / global balance

energy from hinge angles

curved ensemble motion can be "free" except causal limit

A good minimal energy is:

$E = E_{\text{area}} + E_{\text{bend}} + E_{\text{compat}} + E_{\text{boundary}}$ (a) Area term (small dilation allowed) $E_{\text{area}} = \sum_{f\in F} \frac{K_A}{2} \left(\frac{A_f - A_{0,f}}{A_{0,f}}\right)^2 A_{0,f}$

K_A: area stiffness

if you want rigid area quanta, take $K_A \to \infty$

(b) Hinge bending term (main curvature energy) $E_{\text{bend}} = \sum_{e\in E} \frac{K_\theta}{2} \kappa_e^2 L_e$

L_e edge length weight

this is the discrete analog of curvature-squared elasticity / geometric energy

(c) Compatibility/closure constraints

To prevent impossible local shapes, include penalties (or exact constraints):

face closure

edge length consistency

orientation compatibility

discrete Bianchi-like closure (at a vertex/cell)

Symbolically:

$E_{\text{compat}} = \lambda_C \sum(\text{constraint violations})^2$ (d) Boundary / global curvature condition

You mentioned curved boundary conditions. Yes: curvature can be injected by boundary data.

$E_{\text{boundary}} =$ penalty enforcing prescribed boundary curvature/shape

This avoids needing "infinite elements moved" in a finite simulation; the boundary carries the global geometry.

6) Vacuum and "expanding spheres" (Heim-inspired seed state)

Your idea of spheres expanding from an initial state can be formalized as a reference family of embeddings.

Define a vacuum background as a discrete spherical ensemble

At time t, let the reference configuration be approximately a sphere of radius $R(t)$:

faces tiled on a triangulated sphere

reference angles $\theta_{0,e}(t)$ chosen to match that sphere

reference areas $A_{0,f}(t)$ may scale with expansion

Then energy is measured relative to this moving vacuum:

$\kappa_e(t) = \theta_e(t) - \theta_{0,e}(t)$

This is powerful because:

"expansion of spheres" is built into vacuum kinematics

excitations are deviations from expansion, not the expansion itself

You also proposed:

movement of curved ensemble takes no energy, but there is a causality speed limit

That suggests a gauge-like degeneracy:

certain collective deformations are zero modes of E

they carry information/configuration change but no energy cost (or negligible cost)

dynamics constrained by local update speed c

7) Dynamics: causal hinge updates (no continuum fields required)

Define discrete time steps $t_n \to t_{n+1}$.

State variables

At each step:

face bivectors B_f^n

edge curvatures κ_e^n

possibly conjugate momenta p_e^n

Local update rule (causal)

An edge/face can update only using neighbors within one graph step:

update$(e, n+1) = F\big($neighborhood of e at $n\big)$

with a speed bound:

no influence beyond distance $c\,\Delta t$ (converted to graph distance)

This gives your causality rule without assuming standard fields.

Transport of a curved ensemble

Your picture is exactly a soliton-like curvature packet on the hinge network:

front: increase curvature (bend elements)

rear: decrease curvature (relax elements)

packet translates with low net energy change if profile is near a zero mode

This is analogous to defect transport in lattice systems.

8) Why “infinite elements must move” appears — and how to avoid it

You correctly noticed a deep issue:

In a globally constrained rigid network, local bending can force long-range rearrangement.

This is not a bug; it is a compatibility consequence.

Three ways to handle it:

A. Curved boundary condition (your proposal)

Prescribe boundary curvature/shape, solve interior compatibility.

physically: universe/background geometry is not flat infinity

mathematically: well-posed finite-domain problem

B. Expanding vacuum reference

Measure only deviations from a curved/expanding reference $\theta_{0,e}(t)$.

then many “global moves” become vacuum-preserving gauge motions

C. Topological defects

Allow sparse defects/disclinations/dislocations.

localized curvature can then exist without moving everything

these defects may later become particle-like excitations

This is likely the bridge from pure geometry to matter-like structures.

9) Wavefunction as curvature amplitude (your key insight)

This is a very interesting and plausible route.

You suggested:

wavefunction amplitude from curvature

square relates to energy/mass

A clean formulation is:

Curvature complex amplitude

Define a local complex scalar from geometric invariants:

$\psi(x,t) = \alpha\,\kappa(x,t) + i\,\beta\,\pi(x,t)$

where:

κ: coarse-grained curvature amplitude (from hinge angles)

π: conjugate “curvature flux” / rate / phase-like variable

α, β: scaling constants

Then define density:

$\rho(x,t) = |\psi(x,t)|^2$

Interpretations:

$\rho_E \propto |\psi|^2$: local geometric excitation energy density

$\rho_M \propto |\psi|^2/c^2$: effective mass density after coarse-graining

This is not yet quantum mechanics, but it gives the square-law density naturally.

How phase can emerge geometrically

Phase may come from:

orientation/rotation of bivectors

holonomy around loops

discrete connection angle (parallel transport defect)

Then interference-like behavior can emerge when multiple curvature transport paths recombine and phases add before squaring.

That gives a route toward a wavefunction interpretation from geometric holonomy + curvature amplitudes.

10) Minimal formal theory (compact version)

Here is a concise axiomatic version you can iterate on.

Axiom A1 — Discrete geometric substrate

Space at the fundamental scale is a simplicial complex K of oriented triangular area quanta f, each represented by a bivector $B_f \in \Lambda^2(T_xM)$.

Axiom A2 — Rigid plaquettes, flexible hinges

Plaquettes are metrically rigid to first order ($A_f \approx A_{0,f}$), while relative orientation changes across shared edges are dynamical through dihedral angles θ_e.

Axiom A3 — Curvature as hinge deviation

Local curvature excitation is encoded by hinge variables $\kappa_e = \theta_e - \theta_{0,e}$, relative to a reference vacuum geometry (possibly expanding and curved).

Axiom A4 — Geometric energy

Energy is a functional of area strain, hinge curvature, and compatibility constraints:

$E[B,\theta] = \sum_f U_A(A_f) + \sum_e U_\theta(\kappa_e) + \sum_c U_C(\text{closure/compatibility})$ Axiom A5 — Causal local dynamics

The substrate evolves by local update rules with finite propagation speed c, preserving orientation and compatibility constraints up to allowed defect creation rules.

Axiom A6 — Curvature packets and transport

Localized curvature patterns propagate by front-bending/rear-relaxation; translation may occur along near-zero-energy modes while remaining causally bounded.

Axiom A7 — Emergent amplitude

A coarse-grained complex amplitude ψ is defined from curvature and its conjugate geometric flux; $|\psi|^2$ is proportional to local excitation energy density and effective inertial density.

11) What this buys you (important)

This framework gives you, before particles/fields:

a kinematic substrate

a curvature-based energy

a causal update law

a natural square-law density

a path to defects = matter-like excitations

a path to holonomy phase = wave behavior

That matches your goal very well.

12) Immediate next step (concrete and testable)

Build the smallest nontrivial model:

Model M0: 2D triangulated shell in 3D (discrete time)

Triangular mesh approximating a sphere

Fixed edge lengths or stiff edges

Hinge angle energy

Expanding reference radius $R(t)$

Inject one localized curvature packet

Observe transport under local causal rule

Measure:

total energy

packet velocity (bounded)

dispersion

interference of two packets

$|\psi|^2$-like density from curvature and rate

This is the right scale to test your hypothesis before adding any “field” language.

13) Julia / Grassmann.jl / Cartan.jl angle (how to encode)

Even before full simulation, you can define the algebra cleanly.

Grassmann.jl use

represent oriented local basis and bivectors $e_i \wedge e_j$

compute wedge/inner/geometric products

encode orientation and local frame transforms

Cartan.jl use

discrete/coarse differential objects, frame fields, connections (as you scale up)

later: curvature/holonomy in a bundle-like formulation

For the first prototype, you may only need:

a mesh structure

hinge angles

local frames

bivector bookkeeping

Then gradually add Cartan-style connection/holonomy objects.

14) One key caution

To stay disciplined (and make this scientific), keep three layers separate:

Layer 1: exact geometry (definitions, constraints, energy)

Layer 2: emergent phenomenology (packets, wave amplitude, mass density)

Layer 3: Heim correspondence (which parts map to Heim’s original concepts)

This prevents accidental overfitting to interpretations.

If you want, next I can give you a formal M0 specification in math + a Julia prototype skeleton (mesh, hinge curvature, energy, causal update, curvature-to-ψ mapping) in a style compatible with Grassmann.jl.
